Topology 101

~~Obs~~ In the reals ~~Q~~ ($\mathbb{R}$) we have some notion of open and closed intervals

$[-3,5] = \{x \in \mathbb{R} ; 3 \leq x \leq 5\}$

~~open~~ closed

$]-3,5[= \{x \in \mathbb{R} ; 3 < x < 5\}$

~~close~~ open

When we are studying analysis we learn that this notion of open/closed intervals is useful when defining: ~~continuous~~

→ continuous functions
~~→ connectedness of the space~~
→ convergence of sequences
→ convergent subsequences

~~One of the~~ Some of the most common uses of topology is to be able to generalize some ideas like these (and many others) for sets different then the reals

Def: Let X be a set, we define $\mathcal{P}(X)$ as the set of all subsets of X

Ex: Let $X = \{0,1\}$, then $\mathcal{P}(X) = \{\emptyset, \{0\}, \{1\}, \{0,1\}\}$

Obs: For finite sets if $|X| = n$ then $|\mathcal{P}(X)| = 2^n$ (exercise)

For infinite sets $|X| < |\mathcal{P}(X)|$

Def: Given a set X and $\tau \subseteq \mathcal{P}(X)$ we say that (X, τ) is a topological space if

(i) $\emptyset \in \tau$, $X \in \tau$

(ii) if $V_1, \dots, V_n \in \tau$ then $\bigcap_{i=1}^{n} V_i \in \tau$

(iii) if $\{V_i\}_{i \in I} \in \tau$ then $\bigcup_{i \in I} V_i \in \tau$

We say that the elements of τ are the open sets of X.

If V is open in X we say V^c is closed

↳ $V^c = \{x \in X ; x \notin V\}$

Ex: $X = \{0,1\} \Rightarrow \mathcal{P}(X) = \{\emptyset, \{0\}, \{1\}, \{0,1\}\}$

We can have $\tau_1 = \{\emptyset, \{0,1\}\}$ (Trivial)

$\tau_2 = \{\emptyset, \{1\}, \{0,1\}\}$

$\tau_3 = \{\emptyset, \{0\}, \{0,1\}\}$

$\tau_4 = \{\emptyset, \{0\}, \{1\}, \{0,1\}\}$ (discrete)

Ex: Given $X = \mathbb{R}$ we can say that $V \subseteq \mathbb{R}$ (This is the same as saying $V \in \mathcal{P}(X)$) is open if V is an intersection of finitely many open intervals or union of infinitely many open intervals

(in this case the open intervals form a basis of open sets)

Ex: Given X any set we can say $\tau = \{\emptyset, X\}$ (trivial topology) or $\tau = \mathcal{P}(X)$ (discrete topology)

~~Def: A basis for (X,τ) is a subset $B \subseteq \tau$ s.t. every element in τ can be written as union of elements in B~~

Top Grps 101

Def: Let (X, τ_X) and (Y, τ_Y) be two topological spaces and $f: X \longrightarrow Y$. We say that this function is continuous if for every $V \in \tau_Y$ we have $f^{-1}(V) \in \tau_X$.

(pre-image of open is open)

Def: Let $(G, \cdot)$ be a group and let τ be a Hausdorff topology on G. We say that $(G, \tau, \cdot)$ is a Topological group if

$$\cdot : G \times G \longrightarrow G \qquad \text{and} \qquad {}^{-1} : G \longrightarrow G,$$
$$(a,b) \longmapsto a \cdot b \qquad\qquad\qquad a \longmapsto a^{-1}$$

are continuous functions.

Obs: The topology given for $G \times G$ is $\tau_{G\times G} = \{U \times V ; U, V \in \tau_G\}$

Ex$_1$: Let $(\mathbb{R}, +)$ be our group and τ the usual topology on $\mathbb{R}$.

Let $]a,b[$ be an open interval.

$$+^{-1}(]a,b[) = \{(x,y) \in \mathbb{R}^2 ; a < x + y < b\}$$

is open in $\mathbb{R} \times \mathbb{R}$

$-^{-1}(]a,b[) =]-b,-a[$

and because we proved for the basis we also have it is true for the whole top space

Ex 2: Let $(G,\cdot)$ be any group and τ_G the discrete topology

$\Rightarrow$ ~~$G\times G$~~ because every subset of G is open we have that G is a top group (obs: $G\times G$ also will have discrete top)

Def: We say a continuous function $f:(X,\tau_X)\to(Y,\tau_Y)$ is a homeomorphism if it is a bijection, continuous and f^{-1} is also continuous.

Obs: (1) From the def. of top grps we get that $^{-1}: G\to G$ is a homeomorphism

(2) Let G be a top grp and let $g\in G$. Define

$$L_g: G \to G$$
$$x \mapsto gx$$

This map is cont, by the def of top grps, and

$$\cancel{G \to G\times G \to G}$$
$$\cancel{x \mapsto (g,x) \mapsto gx}$$

bijective, ~~for every~~ by the grps axioms. Its inverse is

$$L_{g^{-1}}: G \to G \qquad \text{because}$$
$$x \mapsto g^{-1}x$$

$$(L_{g^{-1}}\circ L_g)(x) = L_{g^{-1}}(gx) = g^{-1}(gx) = (g^{-1}g)x = x$$

~~cont~~: We already know x^{-1} is continuous & L_g is a homeomorphism

(analogous for R_g)

↳ a group looks locally everywhere "the same"

(3) If $H \leq G \Rightarrow H$ is a top group with subspace top.

(4) If G_1, G_2 are grps, $h: G_1 \to G_2$ homom $\Rightarrow h$ is continuous iff it is cont at $1 \in G_1$ (or any other single pt).

More ex of top grps

Ex$_3$: $(\mathbb{C}^*, \cdot)$ and $(\mathbb{R}^*, \cdot)$

Ex$_4$: ~~If $\mathbb{F}$ is a field~~ Let $GL(n, \mathbb{R}) = \{X \in M_{n\times n}(\mathbb{R}) ; \det(X) \neq 0\}$. It is easy to see that this is a subgrp of $\mathbb{R}^{n \times n}$

Cl: This is a top grp with subspace top.

$$\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \right) \xrightarrow{\ \cdot\ } \begin{pmatrix} a_1 a_2 + b_1 c_2 & a_1 b_2 + b_1 d_2 \\ c_1 a_2 + d_1 c_2 & c_1 b_2 + d_1 d_2 \end{pmatrix}$$

polynomial

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \xrightarrow{\ -1\ } (\det A)^{-1} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

polynomial / permutation

Ex$_5$: We can replace in Ex$_4$ $\mathbb{R}$ by any top. field, $\mathbb{Q}_p$ for example

Def: (Basis)

Let (X, τ) top space. $\mathcal{B} \subseteq \tau$ is a basis if every open set $V \in \tau$ can be written as ~~an union~~ $\bigcup_{i \in I} V_i = V$, where $V_i \in \mathcal{B}$

Ex1: ~~Let~~ For any top space τ is a basis

Ex2: Let ~~Q~~ $X = \mathbb{R}^n$ with the ~~usual~~ Euclidean topology. τ in this case is uncountable but one can prove that the set $\mathcal{B} := \{B_r(x) \mid r \in \mathbb{Q}_{\geq 0}, x \in \mathbb{Q}^n\}$ is a basis for (X, τ).

Def: (Basis for an unique Topology)

$\mathcal{B} \subseteq \tau$ is stc. if it satisfies the following properties

(i) $X = \bigcup_{Y \in \mathcal{B}} Y$, (ii) $B_1, B_2 \in \mathcal{B}$ are s.t. $B_1 \cap B_2 \neq \emptyset$ then $\forall x \in B_1 \cap B_2 \; \exists B_3 \in \mathcal{B}$ s.t. $x \in B_3 \subseteq B_1 \cap B_2$

Def: (Subbasis)

Def: